

BPHO ROUND 0 · QUESTION PAPER

Drill bank section 6

25 questions · 60 minutes · single answer, 5 options · **no calculator** · no negative marking

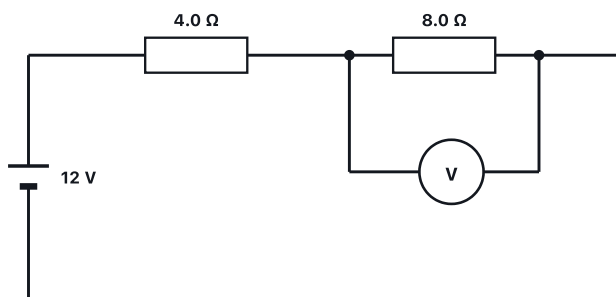
Original questions written for this course, in the Round 0 format. Not BPhO material.

Before you start

- Answer every question. There is no negative marking, so a blank is a wasted mark.
- No calculator. Every answer here is reachable by hand, and the numbers are chosen to divide cleanly.
- Work on paper, then mark yourself against the markscheme PDF.
- The 2025 UK qualifying line was 11 out of 25, so about 44% is the target to beat.

Q1 · A voltmeter of finite resistance loads the potential divider it is measuring, so the reading is below the unloaded value **H**

The cell has negligible internal resistance. The voltmeter has a resistance of $24\ \Omega$ and is connected across the $8.0\ \Omega$ resistor, as the diagram shows.



What does the voltmeter read?

- | | |
|----------------|----------------|
| A 7.2 V | B 8.0 V |
| C 6.0 V | D 4.8 V |
| E 2.4 V | |

Q2 · Propagating a small fractional change through a square root, using the first-order expansion **A**

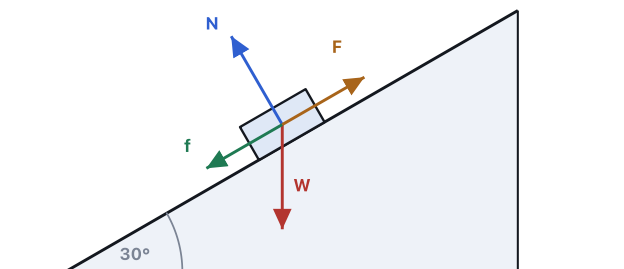
A pendulum is timed and its period is T . Its length is then increased by 4.0 %.

By approximately what percentage does its period change?

- | | |
|-----------------|----------------|
| A 8.0 % | B 4.0 % |
| C 2.0 % | D 1.0 % |
| E 0.50 % | |

Q3 · A block pulled up a rough inclined plane at constant speed: resolving along the plane and getting the direction of friction right **C**

The block has a mass of 2.0 kg and is pulled up the slope at a constant speed by the force F shown. The coefficient of friction between the block and the slope is 0.25, and g is 10 N kg^{-1} .



What is the magnitude of F ?

- | | |
|----------------|---------------|
| A 5.7 N | B 14 N |
| C 15 N | D 24 N |
| E 4.3 N | |

Q4 · A circuit with a nested parallel pair and internal resistance: working from the supply down to the current in one innermost resistor **H**

The cell has an e.m.f. of 12 V and an internal resistance of 2.0Ω . The connecting wires have negligible resistance.

The external circuit is a 5.0Ω resistor in series with two parallel branches. One branch is a single 6.0Ω resistor. The other branch is a 4.0Ω resistor in series with a parallel pair made of 6.0Ω and 3.0Ω .

What is the power dissipated in the 3.0Ω resistor?

- | | |
|------------------|-----------------|
| A 0.24 W | B 1.4 W |
| C 0.080 W | D 0.48 W |
| E 0.16 W | |

Q5 · An order-of-magnitude estimate assembled from a rate, a unit conversion and a lifetime **A**

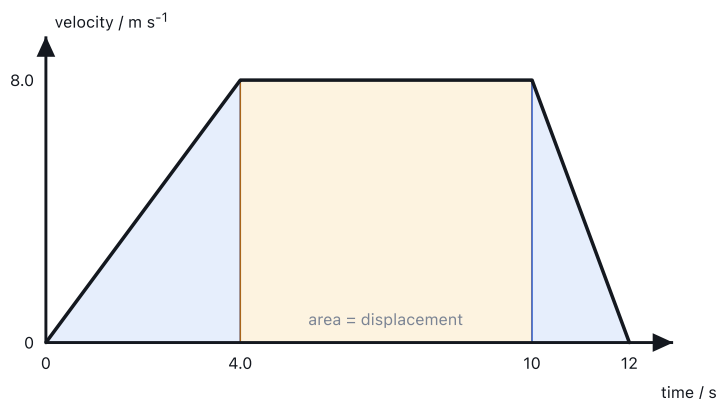
A healthy adult heart beats about 70 times a minute.

Estimate, to the nearest power of ten, how many times a heart beats in a lifetime.

- | | |
|--------------------|-----------------|
| A 10^{11} | B 10^7 |
| C 10^{10} | D 10^8 |
| E 10^9 | |

Q6 · Average velocity from a velocity-time graph: the area under the graph divided by the whole time, expressed as a fraction of the peak speed **B**

The graph shows the velocity of a cyclist over a 12 s interval. The speed rises uniformly from rest to 8.0 m s^{-1} in the first 4.0 s, stays approximately constant at 8.0 m s^{-1} until 10 s, and then falls uniformly to rest at 12 s.

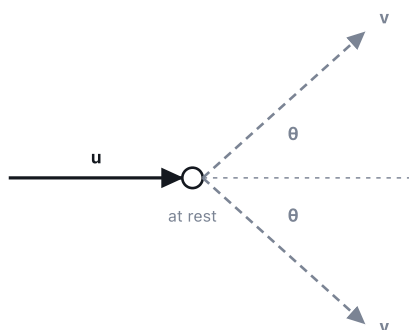


Taking the constant speed as v , what is the cyclist's average velocity over the 12 s?

- A** $v/2$ **B** $3v/4$
C $2v/3$ **D** $5v/6$
E v

Q7 · An elastic collision between equal masses, one at rest, where the two outgoing directions are symmetric: finding the angle **C**

A particle of mass m travelling with speed u strikes an identical particle at rest. The collision is elastic, and the two particles emerge symmetrically, each at the same angle θ to the original direction and each with the same speed v .

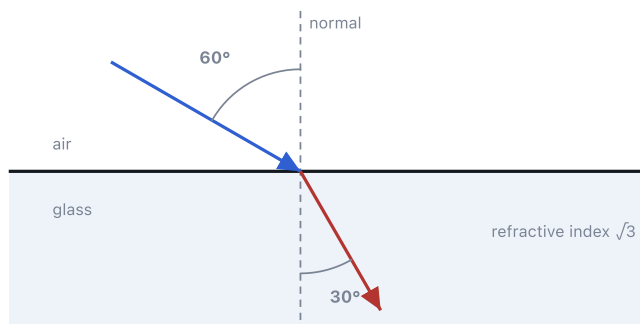


What is θ ?

- A** $\theta = 45^\circ$ **B** $\theta = 60^\circ$
C $\theta = 90^\circ$ **D** $\theta = 30^\circ$
E $\theta = 22.5^\circ$

Q10 · Refraction at a plane surface: finding the speed of light in the material from the two angles, with both measured from the normal **G**

The diagram shows a ray of light travelling in air and meeting the flat surface of a transparent material. The angle of incidence is 60° and the angle of refraction is 30° , both measured from the normal.



What is the speed of light inside the material? Take the speed of light in vacuum as $3.0 \times 10^8 \text{ m s}^{-1}$.

- | | |
|---|---|
| A $2.6 \times 10^8 \text{ m s}^{-1}$ | B $3.0 \times 10^8 \text{ m s}^{-1}$ |
| C $5.2 \times 10^8 \text{ m s}^{-1}$ | D $1.7 \times 10^8 \text{ m s}^{-1}$ |
| E $1.5 \times 10^8 \text{ m s}^{-1}$ | |

Q11 · Power dissipated inside a cell, obtained symbolically by eliminating the current between Ohm's law and the power relation **H**

A cell of e.m.f. $\mathcal{E}$ and internal resistance r is connected to an external resistor of resistance R .

Which expression gives the rate at which energy is dissipated inside the cell?

- | | |
|--------------------------------------|--|
| A $\mathcal{E}^2 r / (R + r)$ | B $\mathcal{E}^2 r / (R + r)^2$ |
| C $\mathcal{E}^2 / (R + r)$ | D $\mathcal{E}^2 R / (R + r)^2$ |
| E $\mathcal{E}^2 r / R^2$ | |

Q12 · The number of nuclei decaying in an interval short compared with the half-life, using the constant-rate approximation **K**

A sample contains 4.0×10^{20} radioactive nuclei. The decay constant of the nuclide is $1.0 \times 10^{-8} \text{ s}^{-1}$.

Approximately how many nuclei decay during the first 60 s?

- | | |
|-------------------------------|-------------------------------|
| A 2.4×10^{14} | B 4.0×10^{12} |
| C 2.4×10^{22} | D 2.4×10^{12} |
| E 2.4×10^{13} | |

Q13 · An estimate of a daily volume: the discriminator is that every conversion from a rate per minute to a total per day must be carried out **A**

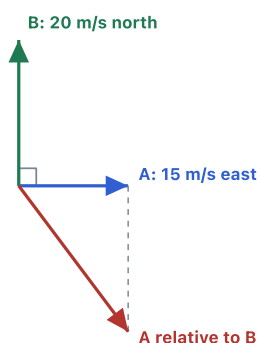
A resting adult breathes in about 0.50 litre of air with each breath, at about 12 breaths a minute.

Estimate, to the nearest power of ten, how many litres of air are breathed in during one day.

- A** 10^1 **B** 10^2
C 10^3 **D** 10^5
E 10^4

Q14 · Relative velocity of two bodies moving at right angles, obtained as a vector difference rather than a scalar one **B**

Two cyclists approach a crossroads on straight roads that meet at right angles. Cyclist P travels due east at 15 m s^{-1} . Cyclist Q travels due north at 20 m s^{-1} .



What is the speed of P relative to Q?

- A** 5.0 m/s **B** 35 m/s
C 25 m/s **D** 18 m/s
E 12 m/s

Q15 · A two-body system released from rest, where the driving mass reaches the floor and the sliding block is then decelerated by friction alone **C**

Block A of mass 3.0 kg rests on a rough horizontal table and is joined by a light inextensible string, passing over a frictionless pulley at the table's edge, to block B of mass 2.0 kg which hangs freely. The coefficient of friction between A and the table is 0.20, and g is 10 N kg^{-1} .

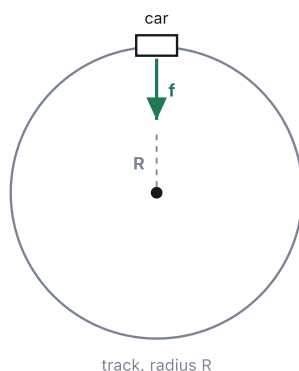
The system is released from rest. Block B falls 1.0 m to the floor, and the string then goes slack.

How much further does block A slide before coming to rest?

- A** 0.70 m **B** 2.8 m
C 1.0 m **D** 1.4 m
E 0.47 m

Q16 · The greatest speed round a flat bend, obtained by equating the friction limit to the centripetal force and giving the answer as an expression **D**

A car of mass m goes round a flat circular bend of radius r . The coefficient of friction between the tyres and the road is μ .



Which expression gives the greatest speed at which the car can take the bend without sliding?

- | | |
|-----------------------------|------------------------------|
| A $\sqrt{(\mu gr)}$ | B μgr |
| C $\sqrt{(2\mu gr)}$ | D $\sqrt{(\mu gr/2)}$ |
| E $2\sqrt{(\mu gr)}$ | |

Q17 · Linear thermal expansion of a rail, using the original length and a temperature difference in kelvin **E**

A steel rail is 2.0 m long at 20 °C. The linear expansivity of steel is $1.2 \times 10^{-5} \text{ K}^{-1}$.

What is the increase in the rail's length when it is heated to 120 °C?

- | | |
|------------------|-----------------|
| A 0.24 mm | B 24 mm |
| C 1.2 mm | D 4.8 mm |
| E 2.4 mm | |

Q18 · The Doppler shift for a source receding from a stationary observer, expanded to first order in the ratio of the speeds **F**

A source of sound of frequency f moves directly away from a stationary observer at a steady speed v . The speed of sound in air is u , and v is much smaller than u .

Which expression gives the frequency heard by the observer, to first order in v/u ?

- | | |
|---------------------------|-----------------------|
| A $f(1 + v/u)$ | B $f(1 - v/u)$ |
| C $fu/(u - v)$ | D $f(1 - u/v)$ |
| E $f(1 + v^2/u^2)$ | |

Q24 · Threshold wavelength for the photoelectric effect, obtained by equating the photon energy to the work function L

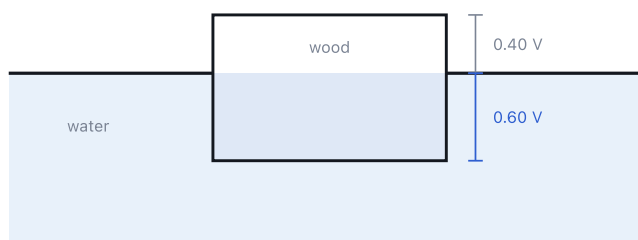
A metal surface has a work function ϕ . Light of wavelength λ falls on it and releases electrons.

Which expression gives the longest wavelength of light that will just release an electron from this surface?

- | | |
|--|--|
| <p>A $\phi/(hc)$</p> <p>C $h/(c\phi)$</p> <p>E hc/ϕ</p> | <p>B $hc\phi$</p> <p>D $c\phi/h$</p> |
|--|--|

Q25 · A floating block: the submerged fraction is the ratio of the block's density to the liquid's, so the mass follows from the displaced volume M

A rectangular block of wood floats in water with 60 % of its volume below the surface, as the diagram shows. The density of water is ρ_w and the block's volume is V .



Which expression gives the mass of the block?

- | | |
|--|---|
| <p>A $\rho_w V$</p> <p>C $\rho_w V/0.60$</p> <p>E $\rho_w V g$</p> | <p>B $0.40 \rho_w V$</p> <p>D $0.60 \rho_w V$</p> |
|--|---|